

(A) Malware Behavior Analysis

1. Introduction

Malware behavior analysis helps in understanding how malicious software interacts with a system. This report covers the analysis of a test malware sample using Cuckoo Sandbox, focusing on file system changes, network activity, and registry modifications.

2. Methodology

A controlled environment was set up using Cuckoo Sandbox to examine the behavior of a known test malware file (**EICAR test file**). The following steps were performed:

Step 1: System Preparation

- Updated and installed necessary dependencies:
- `sudo apt update && sudo apt upgrade -y`
- `sudo apt install python3 python3-pip python3-virtualenv libffi-dev libssl-dev`

`sudo apt install python3-setuptools python3-pyqt5 git mongodb postgresql`

- Configured Cuckoo Sandbox:
- `git clone https://github.com/cuckoosandbox/cuckoo.git`
- `cd cuckoo`
- `python3 -m venv venv`
- `source venv/bin/activate`
- `pip install -U pip setuptools`

`pip install -r requirements.txt`

- Started necessary services:
- `sudo systemctl start mongodb`

`sudo systemctl start postgresql`

Step 2: Malware Analysis

- Downloaded and submitted test malware for analysis:
- `wget https://secure.eicar.org/eicar.com`

`cuckoo submit eicar.com`

- Checked analysis results:
- `cuckoo status`

cuckoo results

3. Findings

- **File System Changes:** Minimal, as the test malware is non-destructive.
- **Network Activity:** No suspicious connections detected.
- **Registry Modifications:** No significant modifications observed.

4. Conclusion

The EICAR test file was successfully analyzed using Cuckoo Sandbox, demonstrating the workflow of malware behavior analysis in a controlled environment.

(B) Digital Forensics Task

1. Introduction

Digital forensics involves analyzing storage media for evidence. This report details forensic examination using Autopsy to analyze a forensic disk image, extract deleted files, and inspect browsing history and logs.

2. Methodology

Step 1: Install and Launch Autopsy

- Installed Autopsy:

```
sudo apt install autopsy -y
```

- Started Autopsy:

```
autopsy
```

- Accessed Autopsy in the browser at: 🖱️ <http://localhost:9999/autopsy>

Step 2: Analyzing the Disk Image

- Created a new case and added a forensic disk image (.e01 format).
- Conducted file recovery and metadata extraction.
- Analyzed logs and browsing history.

3. Findings

Evidence Type	Description
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Deleted Files	Several text and image files were recovered.
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Browsing History Extracted user web activity logs.

Logs Identified system event logs showing past actions.

4. Conclusion

The forensic disk image analysis provided valuable insights, showcasing the ability to recover deleted files and user activity.
